



Data Analytics Project Activity: From Raw Data to Interactive Dashboard



Objective

In this activity, you will take a real-world dataset, clean and prepare it (data wrangling), perform exploratory data analysis (EDA), and finally build an interactive dashboard using Python.

This mirrors a real job workflow used by Data Analysts and Business Intelligence professionals.



Step 1: Dataset Selection (Real-World Data)



Task

Choose **one dataset** from a public source.

Recommended Sources:

- Kaggle: <https://www.kaggle.com/datasets>
 - Google Dataset Search: <https://datasetsearch.research.google.com>
 - Data.gov (Government datasets)
 - World Bank Data
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Suggested Dataset Types (Pick ONE):

- Retail sales data
 - E-commerce transactions
 - Covid-19 cases or health data
 - Airbnb listings
 - Financial stock data
 - Sports performance data
 - Customer churn data
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Requirement

Your dataset must have:

- At least **100+ rows**
 - At least **4 columns**
 - A time-based column (date/month/year) preferred
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Step 2: Data Wrangling (Cleaning & Preparation)

Task

Load your dataset using Python (Pandas) and clean it.

You must perform:

1. Load dataset

```
import pandas as pd
df = pd.read_csv("your_file.csv")
```

2. Handle missing values

- Identify missing values
- Fill or remove them

3. Fix data types

- Convert dates using `pd.to_datetime()`
- Ensure numeric columns are correct

4. Remove duplicates (if any)

5. Create new columns (feature engineering)

Examples:

- Year, Month, Day from Date
 - Profit margin = Profit / Sales
 - Age groups (if demographic data)
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Output Expected:

A clean dataset ready for analysis



Step 3: Exploratory Data Analysis (EDA)



Task

Explore the dataset and answer business questions.



Required Analysis:

1. Basic statistics

- Mean, median, min, max

2. Group analysis

- Sales by category
- Profit by region
- Monthly trends

3. Visualizations (Minimum 4 required)

Use Plotly or Matplotlib:

- Bar chart → Category vs Sales
 - Line chart → Trend over time
 - Pie chart → Distribution
 - Box plot → Spread of values
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Questions to Answer:

- What category generates the highest revenue?
 - What time period shows peak performance?
 - Are there any unusual patterns or outliers?
 - Which region performs best?
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Output Expected:

EDA insights + visual charts

Step 4: Build Interactive Dashboard (Final Project)

Task

Create an interactive dashboard using:

Dash (Plotly)

Dashboard Must Include:

1. Filters

- Year dropdown OR category filter

2. Visualizations (minimum 4–5):

- Bar chart (Category vs Sales)
 - Line chart (Time trend)
 - Pie chart (Distribution)
 - Box plot (Spread)
 - Optional: Sunburst / Heatmap
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Dashboard Features:

- Interactive filtering
 - Dynamic charts
 - Clean layout
 - Clear titles and labels
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Bonus (Optional but recommended):

- KPI cards (Total Sales, Profit, Avg Value)
 - YoY comparison
 - Export dataset feature
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Final Submission Requirements

You must submit:

1. Python notebook or script

- Data cleaning + EDA + dashboard code

2. Dataset used

- Original dataset file or link

3. Short report (1–2 pages)

Include:

- Dataset chosen
 - Key insights
 - Challenges faced
 - What you learned
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Learning Outcome

By completing this project, you will understand:

- Real-world data wrangling
 - Business analysis thinking
 - Data storytelling
 - Building dashboards like industry analysts
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